“What do you think are the key challenges for responsible AI in the next 2-3 years?”

TLDR: While Singapore has a solid foundation with current initiatives and grassroots efforts, the path forward is two-fold towards Responsible AI - on top of us having to move beyond fundamental testing systems towards tackling frontier AI capabilities, we must also double down on existing collaborations as part of our status as a global AI research hub across the Whole-of-Government (WOG) and with international stakeholders.

The primary and most pressing challenge for Responsible AI is in ensuring AI Safety and trustworthiness, but the complexity of the problem has grown far beyond our expectations. The current international landscape for RAI is defined by the urgent technical challenge of AI assurance in existing frontier "black-box" models, which are managed through complex multi-angled analysis techniques and sophisticated control methods towards safe and trustworthy outputs [1].

This global reality establishes the local context, where Singapore's approach to technical AI safety is clearly laid out in the recent Singapore Consensus [2], which proposes a comprehensive, multi-pronged model based on three integrated domains: **Risk Assessment, Development, and Control**. Also, Concordia AI argues while Singapore is demonstrating a greater focus on AI safety as a regional research hub, our current efforts are **focused on local languages and dialects** rather than the development of frontier capabilities; as a result, existing safety efforts are in an early stage and present safeguards techniques are characterized as “**simple**” [3].

Yet, I do acknowledge the strong efforts in the local scene. For one, despite GovTech being known as the primary driver of digital products development and production for public use cases, they have recently been **enhancing efforts in AI safety**, as seen by the RAI team's initiatives through the RAI playbook [4] educating AI practitioners as well as the **conceptualisation and development of new public use cases** [5], complementing existing Whole-of-Government (WOG) efforts like Pair Chat and MAESTRO. On a wider scale, these efforts are also supported by enhanced coordination within and beyond WOG, such as through collaborative platforms like Lorong AI, and industry-academia collaborations like the TRANS Grant research initiative [6].

Given this landscape, the vital question is, 'So what?'; I believe we can tap on the existing AI landscape plus initiatives to **translate existing technical challenges into potential opportunities** to lead as a global research hub in the coming 2-3 years. First, we can double down on our core strengths as a recognized regional research hub with deep human capital and able to engage with both local and international AI communities. We can definitely **foster stronger collaborative efforts** both within and beyond WOG and with academia, building on the aforementioned initiatives to bring researchers working on fundamental, complex problems into the public safety domain.

Second, while existing efforts have concentrated on local, SEA linguistic-based solutions and foundational safeguards, I believe we can extend our goal and aim for **frontier capabilities**, elevating our local status to the international level. This is especially beneficial as existing local directions are **intertwined with fundamental, unresolved issues** of Responsible AI. For example, while the RAI Playbook [4] correctly identifies finetuning as a means toward alignment of AI systems*, this approach faces deep technical challenges as current methods like Reinforcement Learning (including RLHF) do not guarantee inner alignment [7] and may instead mask deception [8]. This failure could then pose as a **public safety risk** across foundational AI models. Additionally, the Playbook [4] highlights model understanding as a primary component as part of the existing approach towards safe deployment of AI**, involving advanced concepts like mechanistic interpretability. However, interpretability itself remains an **open research question** actively being investigated [9], where proposed solutions such as SAEs are still being explored.

While the development of these AI assurance techniques pose as a complex and difficult challenge, I believe these open technical questions present Singapore with a **strategic dual opportunity**. By actively engaging with these issues, we can simultaneously ensure the **safety of our public AI applications** and **solidify our role as a key contributor to global AI safety research** and standards.

** "A common mitigation measure is finetuning or alignment, in which AI models are trained to output human-preferred responses, or aligned to human values, requiring access to model weights."*

*** "At present, our approach to deploying AI models safely involves testing, mitigation and model understanding .. model understanding, whether by understanding the internal mechanisms (i.e., mechanistic interpretability) or outputs (i.e., explainability), is important in increasing transparency of and trust in AI."*

References

[1] https://ai-safety-atlas.com/

[2] https://file.go.gov.sg/sg-consensus-ai-safety.pdf

[3] https://concordia-ai.com/research/state-of-ai-safety-in-singapore/

[4] https://playbooks.aip.gov.sg/responsibleai/responsibleai/

[5] https://huggingface.co/collections/govtech/papers-by-govtech-687606062291f598aa3ecc89

[6] https://www.nrf.gov.sg/rie-ecosystem/ecosystem-wide-fis/snde-fis/

[7] https://www.anthropic.com/research/alignment-faking

[8] https://www.lesswrong.com/w/deceptive-alignment

[9] https://www.lesswrong.com/w/interpretability-ml-and-ai